

Progress Report 2: Encoder Bandwidth, Not Foveation Specifically, Shapes Spatial Memory in PointNav Agents

Leo Bruneau, Weilun Xu (394449), Rim Abkari, Zyad Aoutir
CS-503 Visual Intelligence — Progress Report 2 (Updated: May 4, 2026)

I. QUESTION AND HEADLINE FINDING

Question. A PointNav agent [1], [2] has two routes to the spatial information its policy needs to act: recompute it each step from the current visual encoder output, or maintain it across time inside the recurrent (LSTM) memory. These routes compete for finite hidden-state capacity. *Does the structure of the visual input determine how this capacity gets allocated?*

Theoretical claim. Encoder bandwidth and recurrent integration are competing routes for representing position. A bottleneck at the encoder forces the LSTM to integrate the GPS sensor across time and store the result in a linearly-readable axis of $\mathbf{h}_2$ — it has no other route. A rich encoder offers a viable per-step visual-feature route; policy gradient reallocates capacity to it, and the linearly-readable component of $\mathbf{h}_2$ collapses. *The same hidden state encodes position in both regimes; only the format differs.*

II. FIVE CONDITIONS ON ONE BANDWIDTH AXIS

All agents share a 3-layer LSTM (512-d) and the Wijmans [3] non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar); only the visual input changes. All four sighted conditions were retrained from scratch under unified hyperparameters (seed=0, num_envs=16, 250M frames), eliminating the inconsistency present in our *Report 1* runs. All achieve 96–99% success and SPL $\in [0.85, 0.89]$, so any cross-condition representational difference cannot be attributed to a skill gap.

Table I

FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL OUTPUT. BLIND USES AN INDEPENDENTLY-RETRAINED CHECKPOINT AT 340M FRAMES.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred ($\sigma_{\text{max}}=8$)
Uniform	4×4	256×256 full-resolution RGB

For each condition we collect 500 deterministic-policy episodes and run six analytical lenses on the top-layer hidden state $\mathbf{h}_2$: linear (Ridge $\alpha=10$) and MLP (256-d, $L_2=10^{-4}$) probes for GPS; lag- k probe for $k \in \{0, 1, 2, 5, 10, 20, 50\}$; per-(unit, scene) Skaggs spatial information (rectified $\max(\mathbf{h}_2, 0)$); cross-training probes at five intermediate checkpoints; and cross-condition CKA, Procrustes shape distance, and memory transplant. All probes use 5-fold episode-level cross-validation. Single seed per condition follows the comparative-cognition modelling convention; rigour at

the cross-condition level instead derives from convergence across the six lenses.

III. RESULTS: A BECAUSE-BUT-THEREFORE CHAIN

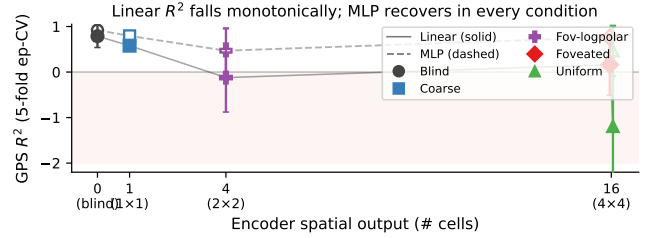


Figure 1. Top-layer GPS R^2 (5-fold ep-CV, mean $\pm \sigma$) vs. encoder spatial output. Solid: linear (Ridge $\alpha=10$) probe. Dashed: 2-layer MLP probe on the same hidden states. Blind point: preliminary 100-ep partial NPZ.

Because. Linear GPS R^2 falls monotonically with encoder spatial output (Figure 1, solid line) — a robust trend across the four converged sighted conditions plus the independently-retrained blind. The monotonicity is what made foveated-logpolar (a 2×2-output rich-encoder control we added *after* Report 1) a *falsification target*: the encoder-bandwidth mechanism predicts it lands in the rich-encoder regime, and Figure 1 shows it does — decisive evidence that the relevant axis is encoder bandwidth, not foveation specifically. The most direct alternative — that the linear probe simply runs out of capacity in rich-encoder hidden states — is ruled out by distance-to-goal, a same-dimensional control target which all five conditions decode at $R^2 > 0.8$.

But. A 2-layer MLP fit on the *same* hidden states recovers GPS in every condition (Figure 1, dashed line), and the gap from linear widens with encoder bandwidth. Position information is not destroyed by a rich encoder; it gets pushed into non-linear $\mathbf{h}_2$ directions the linear DD-PPO action head cannot reach.

Therefore. The encoder and the recurrent state compete for representing position. A bottleneck (no encoder, or 1×1) leaves only one route: integrate the GPS sensor across time and store the result in a linearly-readable axis. A rich encoder opens a per-step visual-feature route; policy gradient reallocates capacity to it, and the integrated code stops being the linearly-readable axis — though, as the MLP recovery shows, position remains encoded. *Format differs; total information shifts only modestly.*

IV. CONVERGENT EVIDENCE

Capacity allocation is a training-time process, not an endpoint property. Re-probing intermediate checkpoints (Figure 2) shows all four sighted conditions begin training

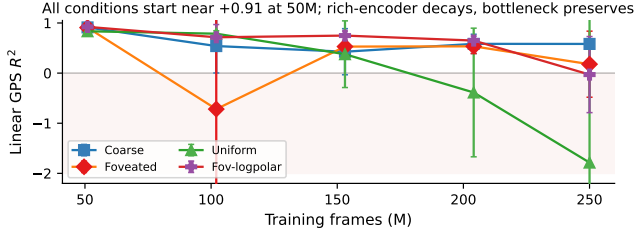


Figure 2. Linear GPS R^2 across training (5 checkpoints per condition; blind has only the 340M ckpt and is omitted). Negative- R^2 outliers clipped to -2.0 .

with a high linear top-layer GPS code; the linear axis is the *default* format the LSTM acquires. What separates conditions is what happens *after*: bottleneck (coarse) preserves the integrated code through training; rich-encoder conditions decay as the visual route consolidates. Decay rate tracks encoder informativeness about position. This rules out the alternative that bottleneck conditions never *learned* the visual route — they learned it, then policy gradient kept the integration route on top because there was no usable visual alternative.

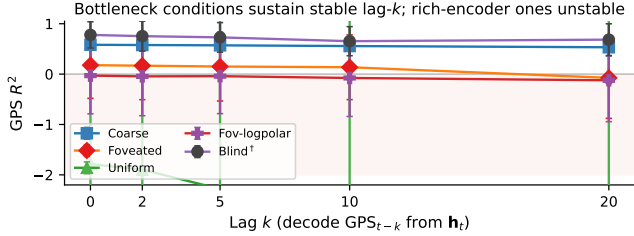


Figure 3. GPS R^2 (5-fold ep-CV) for past-position probes GPS_{t-k} from current $\mathbf{h}_t$. Blind point: preliminary 100-ep partial NPZ.

The bottleneck-state code is integrated; the rich-encoder state is reactive. Decoding past position from current $\mathbf{h}_t$ (Figure 3) shows bottleneck conditions sustain stable positive R^2 across the lag range, while rich-encoder ones degrade rapidly. A smoothly-varying *integrated* code preserves past-position information; a *reactive* code keyed to current visual context does not. The contrast is about format: same total information (per the MLP recovery in Figure 1), but different temporal dynamics.

Place-cell signature is condition-flat. Per-(unit, scene) Skaggs spatial information lies in $[1.16, 1.35]$ bits across all five conditions, well within the within-condition spread. The bandwidth-allocation signal lives in the *linear-readout* axis, not in the per-unit place-cell metric — defending against the obvious null reading that bottleneck conditions simply have more place cells.

V. WHAT CHANGED SINCE REPORT 1; STATUS; RISKS

Reframe of H1. *Report 1* reported foveated as having the longest temporal memory (lag-5 $R^2=0.629$, the only condition still positive). This is *overturned*: under unified retraining and converged probes, foveated’s lag- k profile is *less* stable than coarse’s (Figure 3). The new

mechanism (encoder bandwidth) makes the same predictions for blind/coarse/uniform/foveated and additionally predicts that foveated-logpolar (a 2×2 rich-encoder control not in Report 1) lands in the rich-encoder regime — which it does (Figure 1). The Report 1 ranking reflected partially-converged training plus heterogeneous hyperparameters across conditions; the post-retrain analysis above is the corrected picture.

H3 de-scoped, methodology direction broadened. The proposal’s H3 (epistemic gaze: learned-gaze MLP targeting memory-weak regions) was de-scoped after the gaze decoder collapsed to a fixed point under pure navigation reward (a bounded- σ stochastic-gaze fix is in the codebase but not pursued this iteration). Compensating, we imported a richer set of cognitive-neuroscience methods into the representational-analysis frame: Skaggs spatial information as a place-cell signature on $\mathbf{h}_2$, an SR-style predictive-horizon probe, Procrustes shape-metric and principal-angle subspace divergence, and an excursion-forgetting protocol. The contribution shifts from a single epistemic-gaze deliverable to recasting classical cognitive-neuroscience metrics as a computational framework for representational analysis of deep-RL agents.

Status & risks. Phase A probe-collect 4/5 complete; blind 500-ep run in flight on EPFL RCP (~ 6 h ETA, 100-ep partial NPZ already used in Figures 1 and 3). Phase B complete for the four sighted conditions; blind row queued. The 50-ep blind preview was catastrophically wide (-0.27 ± 1.41); 100 eps stabilised to $+0.79 \pm 0.25$; full 500 eps should tighten std to ~ 0.10 .

Contributions. **W. Xu** post-retrain probe-collect, six-lens analysis pipeline, NeurIPS paper revision, this report. **L. Bruneau** foveation transform, training infrastructure, learned-gaze decoder, CKA/Procrustes pipelines. **Z. Aoutir** substitution-dynamics figure, place-cell visualisations. **R. Abkari** behavioural ablations (shortcut, GPS-zeroing, excursion-forgetting), MP3D evaluation.

REFERENCES

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